

NISHAL STANISLAUS SILVA, *PhD*

Machine Learning Engineer | Computer Vision, Real-Time Inference & Model Serving | PhD

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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PROFESSIONAL SUMMARY

Machine learning engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia, specializing in computer vision models and their path to production. At MAS Holdings I designed and deployed OpenCV-based defect/damage-detection systems on live manufacturing lines – multilingual care-label QC (inspection time cut 99.5%) and loom-side fabric-defect detection (operator headcount cut 50%) – with C++/Python inference engines running on embedded hardware and PLC-integrated automated reject actuation. My postdoctoral research at Trento centered on exactly the problem of optimizing high-latency models for real-time inference: a published detector running at 14 ms on a Raspberry Pi 4, F1 0.76, 74x faster than the DTW baseline it replaced. I bring hands-on PyTorch, OpenCV and TensorFlow experience, backend engineering in AWS (EC2, S3, ECS), containerized services with Docker, and production ETL pipeline design (Airflow); I am extending that foundation into Kafka-based streaming specifically. Canadian Permanent Resident based in Toronto, available immediately, no sponsorship required.

CORE COMPETENCIES

ML Engineering & Model Deployment: computer vision model design and training (PyTorch, OpenCV, TensorFlow), optimizing high-latency models for real-time inference, full model-lifecycle ownership (training, evaluation, deployment, monitoring), visual data annotation pipeline design, evaluation-framework design for complex visual data, containerized model serving (Docker).

Computer Vision & Data Engineering: industrial defect/damage detection with explainable overlays, multilingual visual QC, image registration and warping, PLC-integrated automated reject systems, high-frequency IoT and sensor ETL, backend software engineering in cloud environments (AWS EC2/S3/ECS).

Research & Leadership: benchmark and ablation design, quantified accuracy/latency/CPU trade-off analysis, peer-reviewed publications (JAES, IEEE), mentoring, CI/CD and code-review practice.

TECHNICAL STACK

Python, C++17, SQL, PyTorch, OpenCV, TensorFlow, Keras, scikit-learn, AWS (EC2, S3, ECS, RDS), Docker, Airflow, ETL pipeline design, REST APIs, Git, CI/CD, pytest.

EDUCATION

PhD, Information Engineering and Computer Science – University of Trento, Italy	2021 - 2025
MSc, Telecommunication and Electronic Engineering – Sheffield Hallam University, UK	2016 - 2018
BEng (Hons), Electronic Engineering – Sheffield Hallam University, UK	2013 - 2014

PORTFOLIO

Nebula – open-source C++17 audio-feature-extraction library with per-feature ARM/Raspberry Pi latency benchmarks (github.com/ns2max/nebula).

EXPERIENCE

Independent Researcher | *Self-directed, Toronto, Canada*

Jan 2026 - Present

- Continued applied ML and systems work; two peer-reviewed papers accepted (Smart Drums, JAES; MIRaaS, IEEE IS2 2026), each requiring latency-optimized inference pipelines.

- Used AI-assisted development workflows, including Claude Code, to build and maintain data and tooling pipelines (Nebula; ETL/infrastructure-as-code work on melodiq).

Postdoctoral Researcher | *University of Trento, Trento, Italy*

Jan 2025 - Dec 2025

- Owned the full ML pipeline for streaming audio and sensor data on the EU Horizon EIC Pathfinder project MUS-MET: acquisition, feature engineering, training and evaluation, low-latency edge deployment, and monitoring.
- Optimized a high-latency baseline model for real-time inference: published detector running at 14 ms on a Raspberry Pi 4, F1 0.76, 31.4% CPU utilization – 74x faster than the 1038 ms DTW baseline (JAES 2026).
- Designed frozen-protocol benchmark suites and ablation studies quantifying accuracy/latency/CPU trade-offs, the same rigor needed to validate a vision model before production rollout.

Visiting Researcher | *McGill University, IDMIT/CIRMMT, Montreal, Canada*

May 2024 - Aug 2024

- Built a self-powered embedded sensor system fusing IMU and audio signals via a hierarchical finite-state machine, achieving F1 0.78 on a 500-event corpus (IEEE IS2 2025).

Visiting Researcher | *University of Visual and Performing Arts, Colombo, Sri Lanka*

Jan 2024 - Mar 2024

- Conducted a human-centred evaluation study of a real-time detection system, translating quantitative metrics into usability and adoption recommendations (IEEE I3DA 2025).

Technical Consultant | *Forestpin (Pvt) Ltd, Colombo, Sri Lanka*

Aug 2020 - Feb 2021

- Designed and deployed ML and statistical pipelines for financial forensics and risk detection on large structured enterprise datasets; built SQL + Python backend scoring services running in production.

Research Engineer | *MAS Holdings (Pvt) Ltd, Colombo, Sri Lanka*

Jan 2015 - Jul 2020

- Designed and trained computer vision models for automated defect/damage detection on live production floors: multilingual care-label QC using OpenCV template and feature matching with an explainable overlay, iterated from handheld scanner to industrial camera to a conveyor with PLC-controlled reject, cutting inspection time 99.5%.
- Built a loom-side fabric structural-defect detection system using a microscopic camera array, cutting operator headcount 50%; owned the full model lifecycle from data collection and annotation through deployment and monitoring.
- Ran real-time ingestion and ETL for high-frequency IoT and operational data across manufacturing sites in South Asia, North America, and Africa, backed by C++/Python inference engines on embedded hardware – the closest prior analogue to a Kafka-style streaming pipeline in my track record, though not Kafka itself.
- Introduced CI/CD and code-review practice; mentored engineers and delivered group-wide computer-vision training across the group.

SELECTED PUBLICATIONS (full list: nishal.xyz/publications)

Silva, N. & Turchet, L. (2026). Real-Time Audio Pattern Detection for Smart Musical Instruments. *Journal of the Audio Engineering Society*, 74(3). Silva, N. & Turchet, L. (2026, in press). Smart Drums. *Journal of the Audio Engineering Society*.